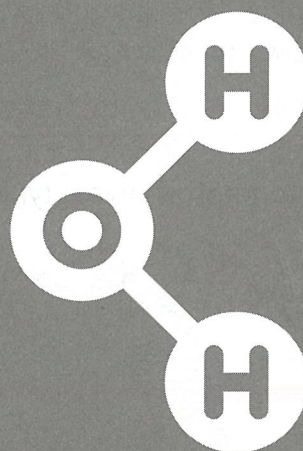




UNSW Global
THE UNIVERSITY OF NEW SOUTH WALES
SYDNEY • AUSTRALIA

**PAPER
A**



2012
ICAS

International Competitions
and Assessments for Schools

SCIENCE

**Educational
Assessment
Australia**

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DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

30 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a 2B or B pencil.

Do **NOT** use a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

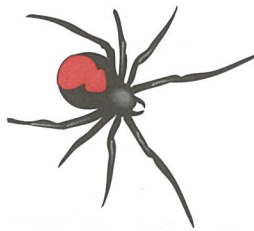
There are **30 MULTIPLE-CHOICE QUESTIONS** (1–30).

Use the information provided to choose the **BEST** answer from the four possible options.

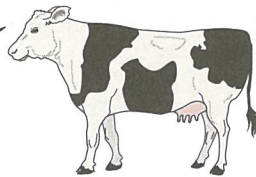
On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a calculator and a ruler.

1. Which two animals have the same number of legs?



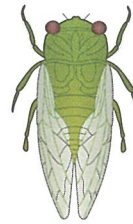
spider



cow



snake



cicada

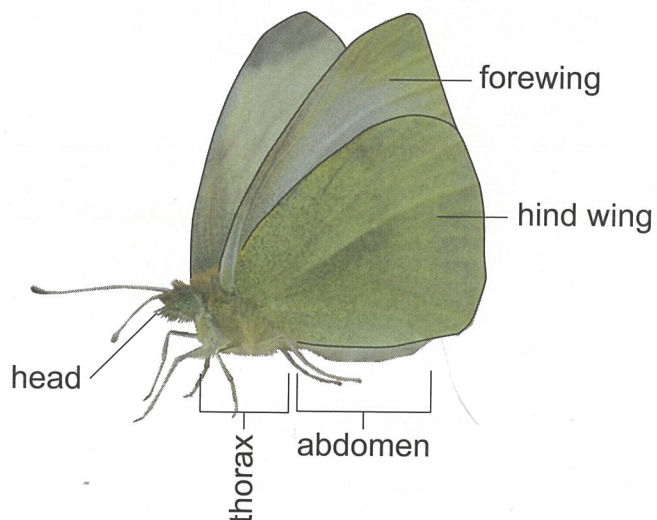


otter

- (A) spider and cow
(C) cow and otter

- (B) snake and otter
(D) spider and cicada

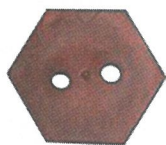
2. The photograph shows a typical butterfly with some of its body parts labelled.



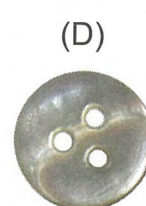
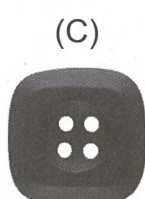
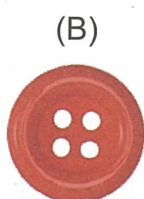
How many wings does a typical butterfly have?

- (A) 2
(B) 4
(C) 6
(D) 8

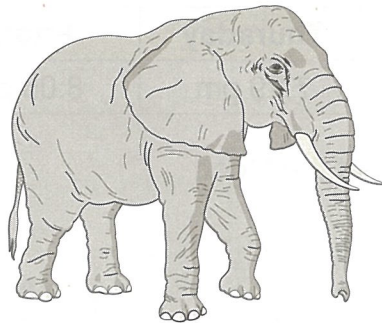
3. The diagram below shows four buttons.



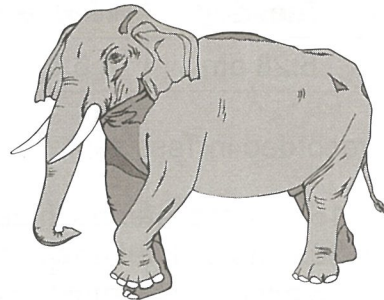
Which of the following belongs with the buttons shown above?



4. The drawing shows two different species of elephant.



African elephant



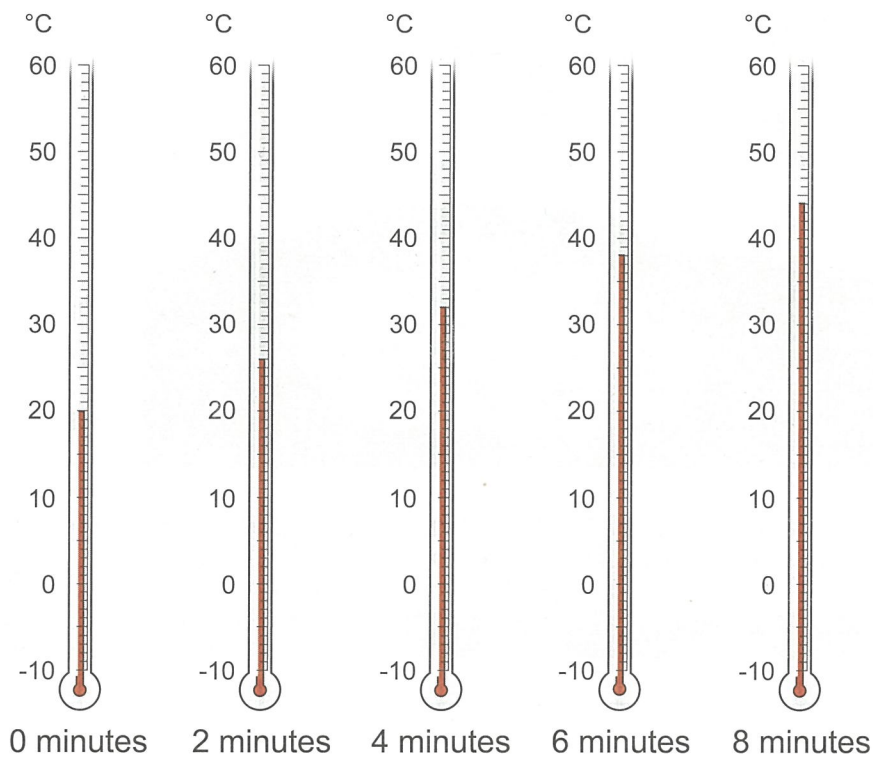
Asian elephant

What part of the African elephant **cannot** be used to tell it apart from the Asian elephant?

- (A) the toes
- (B) the ears
- (C) the eyes
- (D) the trunk

5. Emily heated a saucepan of water for eight minutes. She used a thermometer to measure the temperature of the water every two minutes.

The diagrams show the temperatures during the eight minutes.



What was the change in the temperature of the water after heating for 8 minutes?

- (A) 6 °C
- (B) 18 °C
- (C) 24 °C
- (D) 44 °C

6. Tess watched the Moon and wrote down what time it rose for five days in a row.

Monday	Tuesday	Wednesday	Thursday	Friday
6:01 pm	6:28 pm	6:57 pm	7:29 pm	8:05 pm

The information recorded in Tess' table shows that the Moon

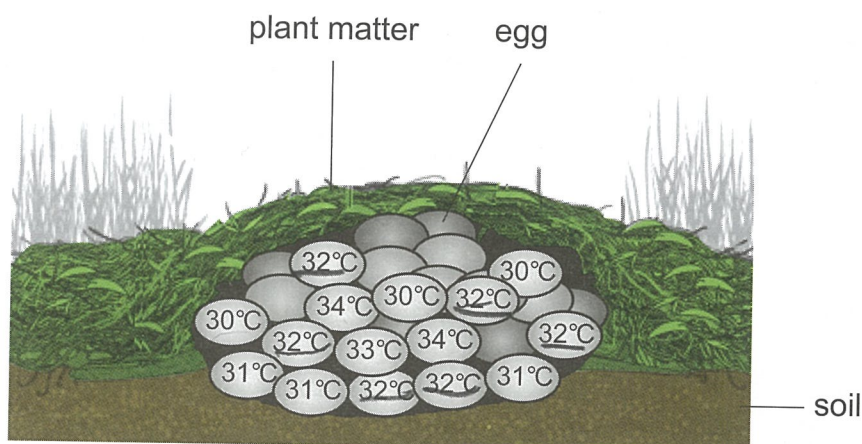
- (A) rose about 15 minutes later each day.
- (B) rose about 30 minutes later each day.
- (C) rose about 15 minutes earlier each day.
- (D) rose about 30 minutes earlier each day.

7. Female saltwater crocodiles lay their eggs in a nest in the soil and then cover the eggs with plant matter.

The temperature of each egg determines whether the egg will hatch a male crocodile or a female crocodile.

- Eggs with a temperature below 32 °C will always hatch female crocodiles.
- Eggs with a temperature of 32 °C will always hatch male crocodiles.
- Eggs with a temperature above 32 °C will always hatch female crocodiles.

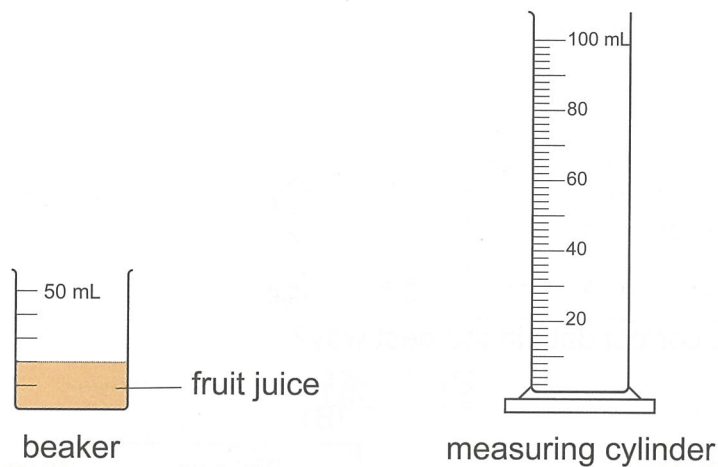
The diagram below shows crocodile eggs in a nest. The temperature of some eggs is marked on them.



How many of the marked eggs will hatch as male crocodiles, and how many of the marked eggs will hatch as female crocodiles?

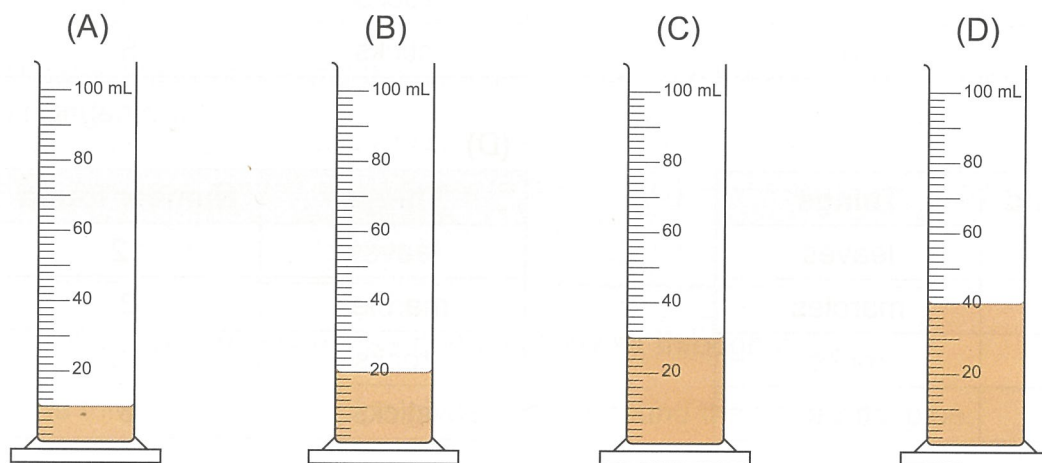
	Male crocodiles	Female crocodiles
(A)	6	9
(B)	6	6
(C)	9	6
(D)	12	3

8. Ben has a beaker and a measuring cylinder. The beaker has some fruit juice in it.



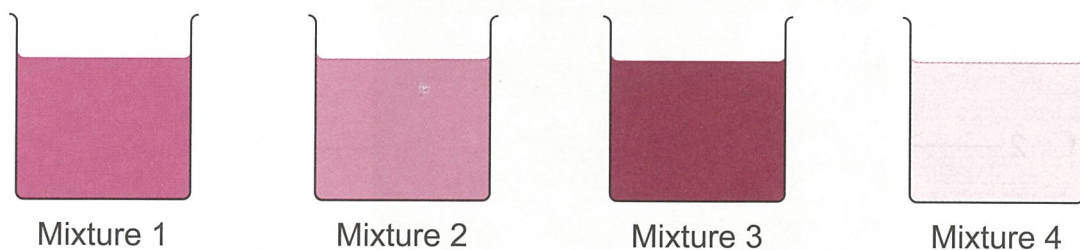
Ben pours all the fruit juice from the beaker into the measuring cylinder.

Which diagram shows the level of the fruit juice in the measuring cylinder?



9. Emma made four mixtures by adding different amounts of food colouring to water. She used the same type of food colouring in every mixture.

The diagrams show the four mixtures she made.



Which mixture has the greatest amount of food colouring in it?

- (A) Mixture 1 (B) Mixture 2 (C) Mixture 3 (D) Mixture 4

10. A group of students made a tally of the things they found in a sandpit.

leaves	
marbles	
rocks	
sticks	()

Which option presents the correct data in the best way?

(A)

Number found	Things
leaves	12
marbles	3
rocks	16
sticks	8

(B)

Things	Number found
leaves	16
marbles	3
rocks	12
sticks	8

(C)

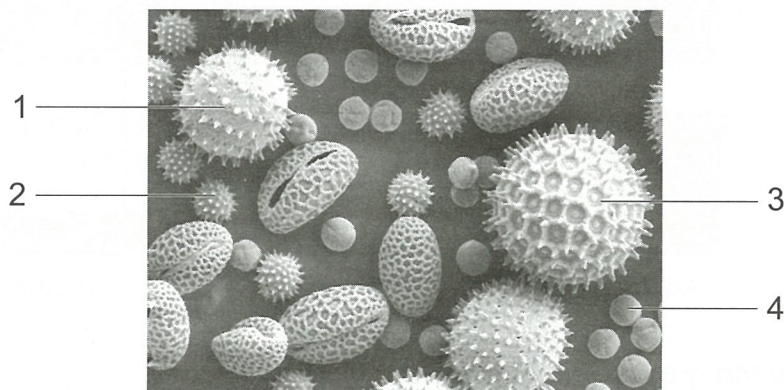
Number found	Things
12	leaves
3	marbles
16	rocks
8	sticks

(D)

Things	Number found
leaves	12
marbles	3
rocks	16
sticks	8

11. Flowers make pollen grains.

The photograph shows different pollen grains as seen through a microscope.

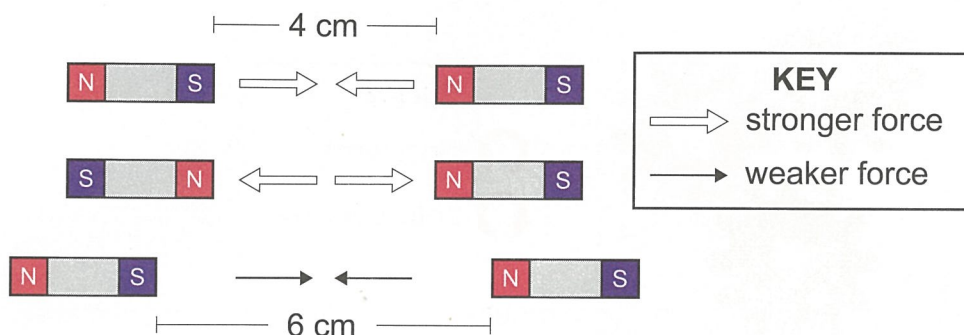


Which list gives the pollen grains in order of size, from smallest to largest?

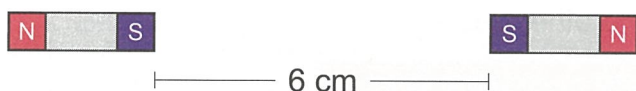
- (A) 3, 1, 2, 4 (B) 3, 1, 4, 2 (C) 4, 2, 1, 3 (D) 4, 2, 3, 1

12. The diagrams show the same two magnets arranged in three different ways.

The arrows show the strength and direction of the force between the magnets for each arrangement.



Jane then arranged the two magnets as shown below.



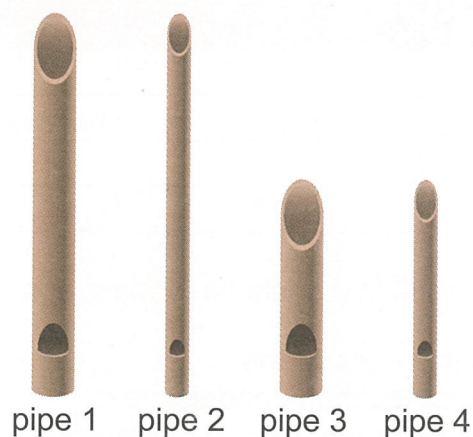
What is the strength and the direction of the force between the two magnets for this arrangement?

- (A) ———→ ———← (B) ———← ———→ (C) ———→ ———← (D) ———← ———→

13. Pipes can produce sounds when you blow air through them.

Jane thinks that a wide pipe gives a lower sound than a narrow pipe.

Jane has the four pipes shown below.

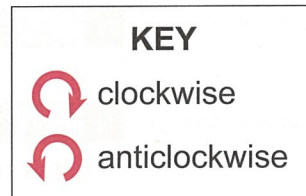
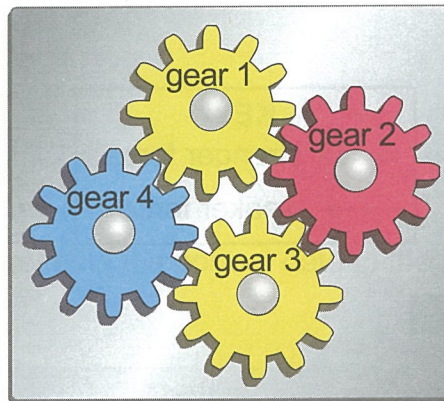


Through which two pipes should Jane blow air to find out if her idea is correct?

- (A) pipe 1 and pipe 2 (B) pipe 1 and pipe 3
 (C) pipe 2 and pipe 3 (D) pipe 2 and pipe 4

14. Gears are wheels with teeth.

The diagram shows four gears.



Gear 1 is rotating clockwise.

What is the direction of rotation of each of the other gears?

	Gear 2	Gear 3	Gear 4
(A)	clockwise	anticlockwise	clockwise
(B)	anticlockwise	clockwise	anticlockwise
(C)	anticlockwise	anticlockwise	clockwise
(D)	clockwise	clockwise	anticlockwise

15. Michelle counted how many weeds of different types she found in four areas of a nature reserve.

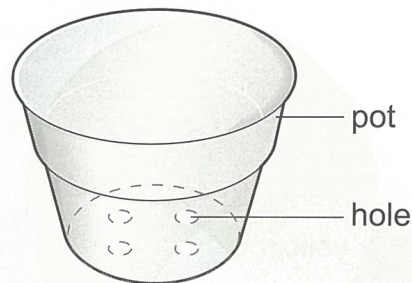
The table shows her results.

Area	Number of each weed		
	Dandelions	Thistles	Blackberries
1	5	7	0
2	3	7	9
3	6	3	13
4	0	12	0

Which statement about the weeds that Michelle found is correct?

- (A) She found dandelions in all four areas.
- (B) She found more blackberries than dandelions.
- (C) She found blackberries wherever there were thistles.
- (D) She found more thistles than dandelions in each area.

16. Jane wanted to find out which type of soil holds the largest amount of water. She used four identical pots like the one shown. Each pot had four holes in the bottom.



Jane filled each pot with an equal amount of soil. Each pot had a different type of soil. Then she added the same amount of water to each pot. Some water leaked out of each pot.

How can Jane find out which type of soil holds the largest amount of water?

- (A) measure how much water comes out of each pot
- (B) measure the amount of soil falling from each pot
- (C) test which soil loses water first
- (D) test which soil feels wet

17. The diagrams show four tiles, each with part of a plant drawn on it.



tile 1



tile 2



tile 3

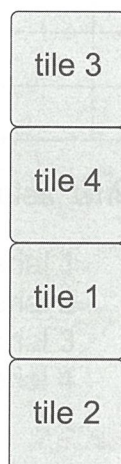


tile 4

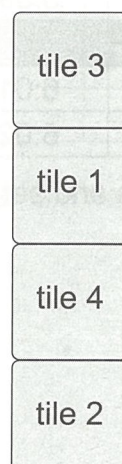
When the tiles are lined up correctly, the drawings show the complete plant.

How do the tiles have to be lined up to show the complete plant?

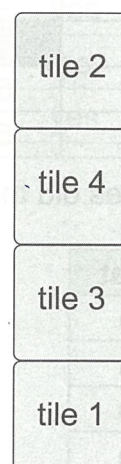
(A)



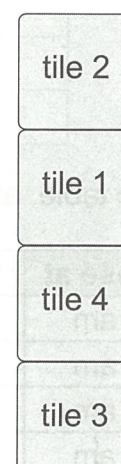
(B)



(C)

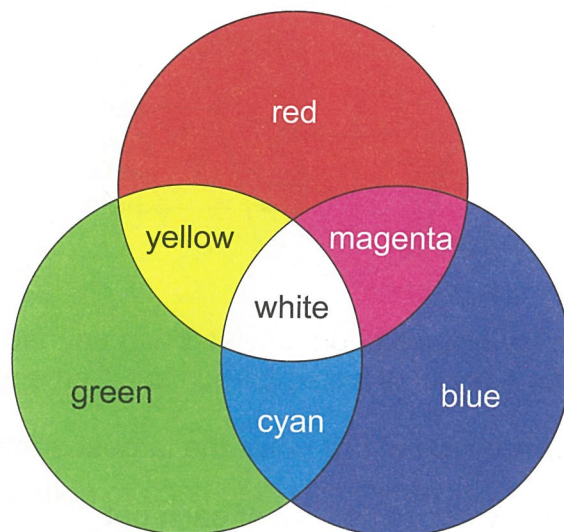


(D)



For questions 18 and 19 use the information below.

The diagram shows what happens when you shine red, green and blue light onto white paper.



18. What colour light is produced where red light overlaps only with green light?
- (A) cyan (B) magenta (C) yellow (D) white
19. Which colours of light produce white light when they overlap?
- (A) red and blue (B) red and magenta
(C) green and cyan (D) green and magenta

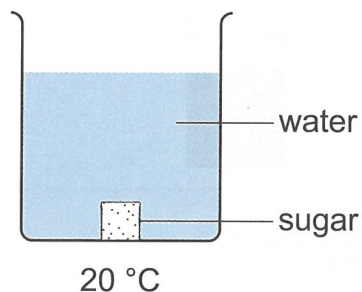
20. The table shows the times that the Sun rose and set during one week. The information for day 5 is missing.

Day	Sun rose at	Sun set at
1	5:50 am	5:56 pm
2	5:48 am	5:57 pm
3	5:46 am	5:58 pm
4	5:44 am	5:59 pm
5		
6	5:40 am	6:01 pm
7	5:38 am	6:02 pm

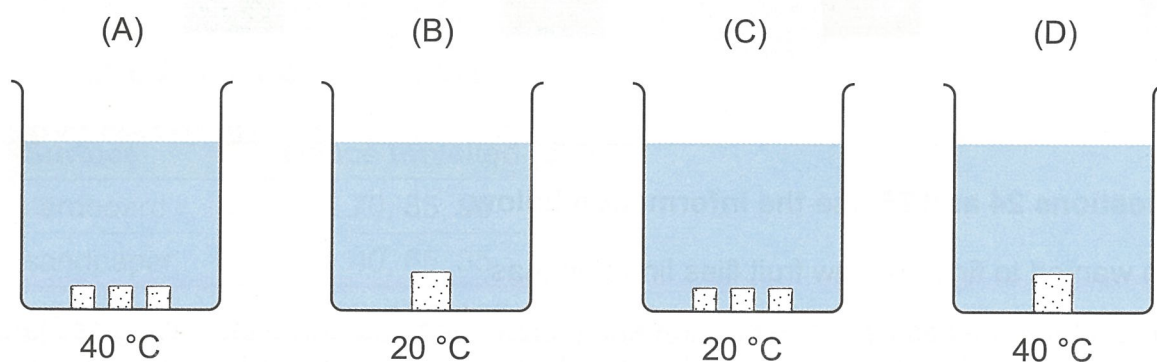
Based on the table, at what times did the Sun rise and set on day 5?

	Sun rose at	Sun set at
(A)	5:42 am	6:00 pm
(B)	5:42 am	5:58 pm
(C)	5:46 am	6:00 pm
(D)	5:46 am	5:58 pm

21. Linda wanted to know if sugar dissolves faster in warm water than in cold water. This is how she set up the first part of her experiment.



Which set-up should Linda use to correctly complete her experiment?



22. Peter wanted to find out if some materials were metals.

He used two rules:

- Metals must be shiny and not dull.
- Metals can be bent into a new shape and stay in that new shape.

Here are Peter's results for four materials.

Material	Shiny	Bend and stay in new shape
1	no	no
2	yes	yes
3	yes	no
4	no	yes

Using the rules, which material is a metal?

- (A) material 1
(B) material 2
(C) material 3
(D) material 4

23. These photographs show what the Moon looked like on some nights during September. The photographs were taken from the same place on Earth, on the dates shown.



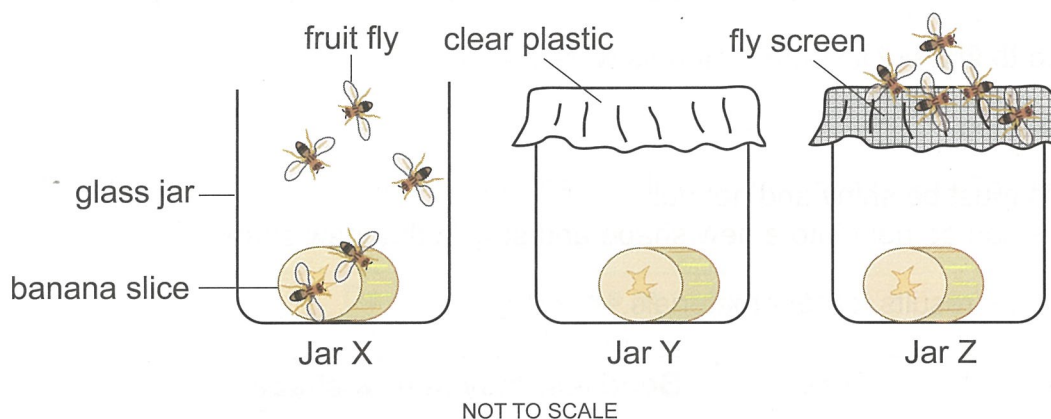
What would the Moon have looked like on September 26?



For questions 24 and 25 use the information below.

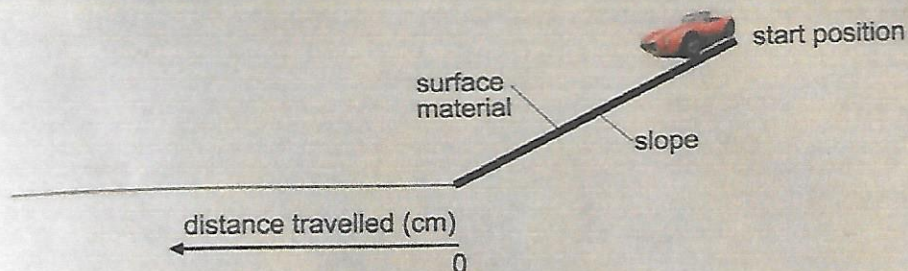
Nathan wanted to find out how fruit flies find bananas.

Nathan cut three pieces from one banana and placed them into three identical glass jars. He covered two of the jars with different materials. He placed all three jars together on a table outside. The diagrams show what Nathan observed a short time later.



24. Based on these observations, how do fruit flies find bananas?
- (A) by sight (B) by smell
(C) by either sight or smell (D) by smell and taste
25. Which condition did Nathan have to keep the same for each jar to make his investigation a fair test?
- (A) the type of cover (B) the type of fruit fly
(C) the number of fruit flies (D) the size of the banana slice

26. Georgie wanted to find the distance a toy car would travel after rolling down a slope.



She used three different surface materials on the slope. She rolled the car 3 times on each surface.

This table shows Georgie's results.

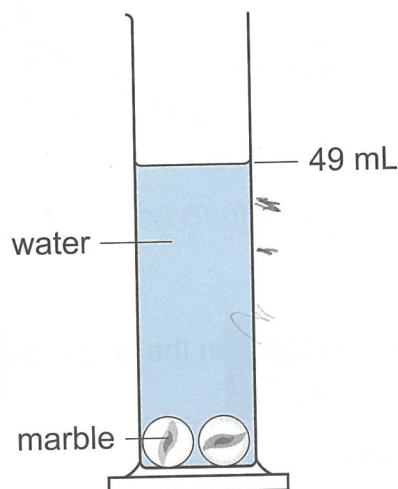
Surface	Distance travelled (cm)
cardboard	70, 85, 95
sandpaper	40, 65, 35
plastic	90, 110, 125

What can Georgie say about her results?

- (A) The surface of the slope did not make any difference to how far the car travelled.
- (B) The car always travelled further after the sandpaper slope than the plastic slope.
- (C) The car always travelled further after the plastic slope than the cardboard slope.
- (D) The car always travelled further after the cardboard slope than the sandpaper slope.

27. Joel had a measuring cylinder with some water in it.

He added two identical marbles and the water level rose to 49 mL.



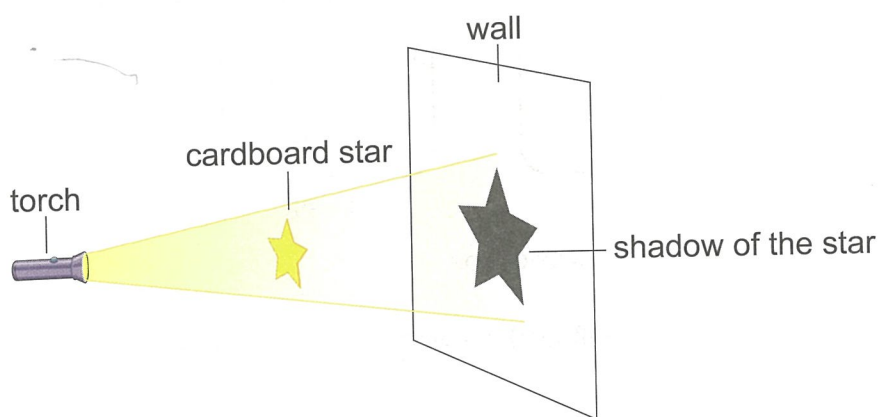
Joel then added a third identical marble and the water level rose to 51 mL.

How much water did the cylinder contain just before Joel added the marbles?

- (A) 43 mL (B) 45 mL (C) 47 mL (D) 49 mL

28. Mary wanted to see if the shadow of a cardboard star changes when she moves the torch closer to the cardboard star.

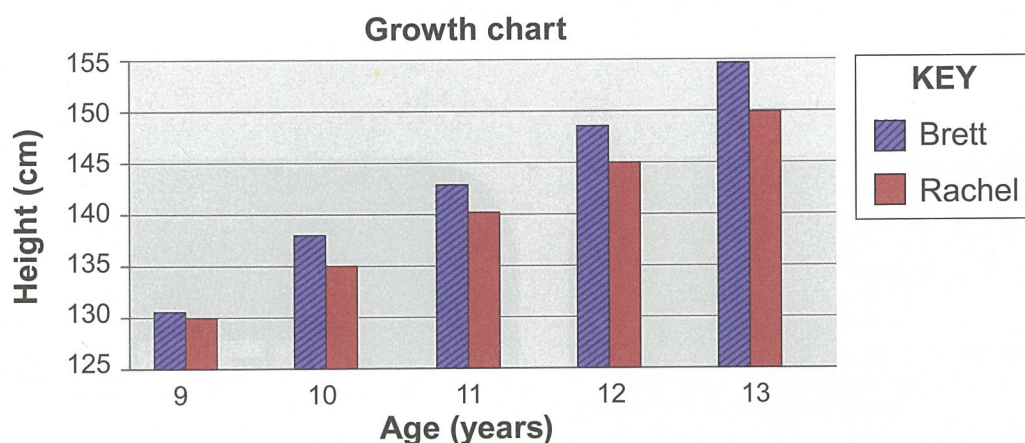
She shone the light from the torch onto the cardboard star in a dark room.



What should Mary change and what should she keep the same in her investigation?

	Change the distance between	Keep the same distance between
(A)	the cardboard star and the torch	the torch and the wall
(B)	the cardboard star and the torch	the cardboard star and the wall
(C)	the cardboard star and the wall	the torch and the wall
(D)	the cardboard star and the wall	the cardboard star and the torch

29. Brett and Rachel had their heights measured on their birthdays over a number of years. The growth chart shows this information.



What conclusion can be drawn from the information in the growth chart?

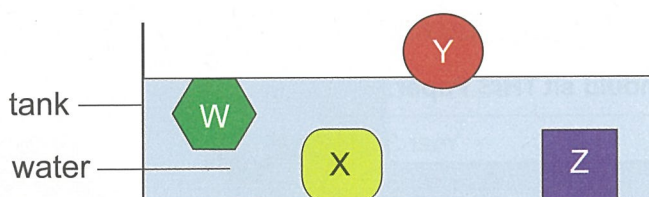
- (A) Brett is older than Rachel.
- (B) Brett and Rachel have the same birthday.
- (C) Brett was taller than Rachel on his 10th birthday.
- (D) Brett grew faster than Rachel between the ages of 9 and 13.

30. Objects either sink or float when placed in water.

An object will sink in water if the mass of the object is greater than the mass of water with identical volume to that of the object.

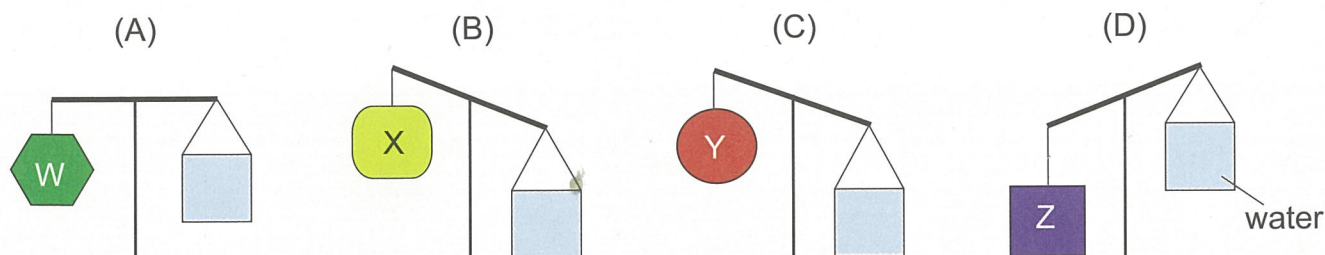
Objects W, X, Y and Z have identical volumes.

The diagram shows what happened when the objects were placed into a tank of water.



The diagrams below show what happened when the mass of each object was compared to the mass of an identical volume of water.

Which diagram is **not** correct?





S

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Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Paper G	Paper H	Papers I & J
1	C	B	B	D	C	C	B	B	D
2	B	B	D	B	B	D	A	C	C
3	A	A	D	A	D	B	C	B	C
4	C	C	C	B	C	B	A	C	C
5	C	D	B	C	B	C	C	B	D
6	B	C	D	C	C	C	C	A	C
7	A	A	C	D	B	B	B	A	B
8	B	A	B	B	D	A	D	A	A
9	C	B	B	B	B	C	C	D	A
10	D	C	C	A	C	B	A	D	D
11	C	D	D	A	B	D	B	C	D
12	D	D	C	A	C	C	B	C	D
13	A	C	D	B	B	D	A	B	B
14	B	A	C	D	D	D	B	A	C
15	B	A	B	A	D	C	B	C	C
16	A	B	A	A	B	D	C	D	B
17	B	A	B	C	C	B	D	C	C
18	C	A	A	B	A	B	A	B	C
19	D	C	A	B	C	B	A	A	B
20	A	B	B	C	C	A	D	C	B
21	D	D	C	A	A	A	B	B	A
22	B	B	B	B	A	A	C	D	D
23	D	D	A	D	C	D	D	C	D
24	B	B	D	B	C	D	C	A	B
25	D	B	C	D	A	C	D	D	D
26	D	A	B	C	A	D	B	D	D
27	B	D	C	A	C	A	B	D	D
28	B	C	B	C	A	D	B	C	B
29	D	D	B	C	B	B	D	D	B
30	B	D	B	B	C	B	C	D	C

(Please turn over)



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Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Paper G	Paper H	Papers I & J
31			A	C	B	D	D	D	D
32			C	D	D	D	D	D	D
33			D	D	D	A	B	C	A
34			C	B	D	B	C	C	C
35			D	B	B	D	C	B	D
36			B	B	B	C	C	A	B
37			D	C	D	A	B	B	D
38			D	D	B	D	D	C	D
39			A	A	B	C	B	D	C
40			A	D	D	B	C	A	C
41					D	A	C	D	C
42					D	C	D	D	B
43					D	C	D	C	C
44					A	D	D	A	D
45					D	C	D	B	D

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